

SIMATS ENGINEERING
ASSESSMENT TOOL II – REVERSE ENGINEERING TASK (MEDIUM)

Course Name: Ethical Hacking for Secure Networks	Code: ITA1436	CO: CO2
Duration: 90 min Date:	Total Marks:	Reg. No / Name : 192325076 / SHAM S.

CO2: *Execute footprinting in social engineering such as DNS analysis, competitive intelligence gathering, and attack simulations to collect relevant information for identifying potential security risks. (BL3)*

Assessment Tool Weightage: 30%

Instructions: Answer all questions. Each question carries 10 marks.

Task: 1 A cybersecurity team discovers a suspicious executable named **FootPrintX.exe** on a penetration tester's workstation. The application appears to collect information about target organizations, including domain names, IP addresses, and network details. The source code is unavailable, and the analyst must reverse engineer the binary to understand its functionality. Your task is to identify how the tool performs footprinting activities.

Questions:

Question 1: Which functions or modules are responsible for gathering target information?

Answer:

The modules responsible for gathering target information are:

- DNS Lookup module
- WHOIS Lookup module
- IP Address Resolution function
- Network Scanner module
- Host Enumeration function
- Operating System Identification module

These modules collect information such as domain names, IP addresses, hostnames, and network details.

Question 2: What footprinting techniques are implemented in the executable?

Answer:

The executable performs several footprinting techniques, including:

- DNS enumeration
- WHOIS information gathering

- IP address lookup
- Host discovery (Ping Sweep)
- Port scanning
- Network enumeration
- Operating system fingerprinting

These techniques help attackers or penetration testers collect information about the target organization.

Question 3: How can the information collection process be disabled or modified?

Answer:

The information collection process can be prevented by:

- Removing or patching the footprinting functions in the executable.
- Blocking DNS and WHOIS requests using firewall rules.
- Restricting network access.
- Disabling unnecessary network scanning modules.
- Applying executable integrity monitoring or endpoint protection.

Task: 2 A file named **DNSScan.exe** has been found during a security audit. Preliminary analysis suggests that the program attempts to gather DNS information from external servers. Security analysts suspect that it performs DNS zone transfer operations to obtain detailed records of a target domain. Reverse engineer the executable to determine its behavior and associated risks.

Questions:

Question 1: Which function is responsible for initiating DNS requests?

Answer:

The DNS request module is responsible for initiating DNS queries. During reverse engineering, functions such as **DnsQuery()**, **gethostbyname()**, or similar Windows networking APIs can be identified as the components responsible for DNS communication.

Question 2: Does the application attempt an AXFR (Zone Transfer) request? Justify your answer.

Answer:

Yes. The executable appears to perform an **AXFR (DNS Zone Transfer)** request because it

attempts to retrieve all DNS records from the target DNS server. This behavior indicates that the program is trying to obtain the complete DNS zone database.

Question 3: What information can an attacker obtain using the extracted DNS records?

Answer:

An attacker can obtain:

- Hostnames
- Internal and external IP addresses
- Mail server (MX) records
- Name server (NS) records
- CNAME aliases
- Subdomains
- Network infrastructure details

This information assists in planning further attacks.

Task: 3 An organization identifies a tool called **IntelGather.exe** that collects publicly available information from websites and online repositories. The application is suspected of gathering employee details, email addresses, and technology information related to organizations. Since the source code is unavailable, reverse engineering is required to determine the data collection mechanisms. Analyze the executable and identify its intelligence-gathering capabilities.

Questions:

Question 1: What types of information are collected by the application?

Answer:

The application collects:

- Employee names
- Email addresses
- Phone numbers
- Company information
- Public documents
- Technology stack
- Domain details
- Social media information

Question 2: Which external websites or services are accessed during execution?

Answer:

The application may access:

- Company websites
- LinkedIn
- GitHub
- WHOIS databases
- DNS lookup services
- Search engines
- Public repositories
- Social media platforms

Question 3: How can the collected intelligence be misused by attackers?

Answer:

Attackers can use the information for:

- Spear phishing
- Social engineering
- Password guessing attacks
- Identity theft
- Credential harvesting
- Reconnaissance before network attacks
- Targeted cyber attacks

Task: 4 A suspicious program named **MailCraft.exe** is discovered on a compromised system. During execution, the application automatically creates convincing email messages that imitate legitimate organizations. Analysts suspect that the tool is used to perform phishing attacks against employees. Reverse engineer the executable to identify how the emails are generated and customized.

Questions:

Question 1: Which functions are responsible for generating phishing emails?

Answer:

The responsible functions include:

- Email template generation module
 - SMTP sending module
 - Message customization function
 - HTML email generation module
 - Attachment handling function
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Question 2: What information is used to personalize the messages?

Answer:

The application uses:

- Employee names
- Company names
- Email addresses
- Job titles
- Department names
- Organization logos
- Previous communication details

This makes phishing emails appear legitimate.

Question 3: How can the phishing functionality be prevented or removed?

Answer:

The phishing functionality can be prevented by:

- Removing the email generation module.
- Blocking SMTP communication.
- Using antivirus or endpoint detection tools.
- Applying application whitelisting.
- Monitoring outbound email traffic.
- Educating users to recognize phishing attempts.

Task: 5 A binary file named **PortalClone.exe** imitates the login page of a popular online service. Users who enter their credentials unknowingly submit their information to the application. The security team needs to understand how the credentials are captured, stored, and transmitted. Reverse engineer the program to identify its malicious behavior.

Questions:

Question 1: How does the application capture usernames and passwords?

Answer:

The application captures credentials by:

- Displaying a fake login page.
 - Recording user input entered into the login fields.
 - Intercepting keyboard input.
 - Storing submitted usernames and passwords before forwarding or saving them.
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Question 2: Where are the captured credentials stored or transmitted?

Answer:

Captured credentials may be:

- Stored in local files.
 - Saved in an internal database.
 - Written to log files.
 - Sent to a remote command-and-control (C2) server through HTTP, HTTPS, or SMTP.
 - Uploaded to an attacker-controlled server.
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Question 3: What security risks arise from the use of such applications?**Answer:**

The major security risks include:

- Credential theft
- Unauthorized account access
- Identity theft
- Financial fraud
- Data breaches
- Privilege escalation
- Organizational compromise
- Loss of confidential information

Code of Conduct:

I SHAM S. (192325076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of System	20%				
Decomposition	25%				
Identification of Components	20%				
Reconstruction	20%				

Documentation		15%				
Criteria	Understanding of System	Decomposition	Identification of Components	Reconstruction	Documentation	Total
Q1						
Q2						
Q3						
Q4						
Q5						